CONSULTATION REASON: Acute onset pleuritic right-sided chest pain, tachycardia, and hypoxemia following a four-hour transcontinental flight.

HISTORY: The patient is a 52-year-old male with no prior cardiopulmonary history who underwent arthroscopic left knee meniscus repair three weeks ago. He ambulated minimally during a long flight yesterday. This morning, he awakened with sudden sharp right-sided lateral chest pain aggravated by deep inspiration and associated with moderate dyspnea. He denies cough, hemoptysis, lower extremity swelling, or prior thromboembolism.

PHYSICAL ASSESSMENT:

General: Alert, anxious, in mild respiratory distress speaking in shortened sentences.

Vital Signs: Heart rate 112 beats per minute (sinus tachycardia), blood pressure 118/74 mmHg, respiratory rate 26 breaths per minute, temperature 37.1 degrees Celsius, SpO2 90% on room air improving to 96% on 2 liters nasal cannula oxygen.

Cardiovascular: Tachycardic, regular rhythm. Normal S1, loud P2 sound at the left upper sternal border. No S3 or S4 gallop. No jugular venous distension.

Chest: Clear to auscultation bilaterally without wheezing, rales, or rhonchi. Right chest wall is non-tender to direct palpation.

Extremities: Left calf circumference is 1.5 cm greater than right calf; mild calf tenderness upon deep compression without palpable cords.

DIAGNOSTIC DATA:

Electrocardiogram: Sinus tachycardia at 110 bpm with S1Q3T3 pattern (deep S wave in lead I, Q wave and inverted T wave in lead III) and new incomplete right bundle branch block.

D-Dimer: Markedly elevated at 3,840 ng/mL FEU (reference < 500 ng/mL).

Troponin I: Borderline elevated at 0.08 ng/mL (reference < 0.04 ng/mL).

NT-proBNP: Elevated at 920 pg/mL.

Computed Tomography Pulmonary Angiography (CTPA): Demonstrates acute saddle pulmonary thromboembolism with occlusive emboli in the right main pulmonary artery extending into the right interlobar, right middle, and right lower lobe segmental branches, along with non-occlusive filling defects in the left lower lobe pulmonary arteries. Right ventricular-to-left ventricular diameter ratio is 1.2 on axial reformatted CT slices, indicating acute right ventricular strain without pulmonary infarction.

Bedside Echocardiogram: Demonstrates right ventricular hypokinesis with sparing of the RV apex (McConnell sign) and mild tricuspid regurgitation.

ASSESSMENT AND MANAGEMENT:

1. Acute intermediate-high risk (submassive) pulmonary thromboembolism with documented biomarker positivity and right ventricular strain in a hemodynamically stable patient.

2. Initiate therapeutic anticoagulation immediately with intravenous unfractionated heparin bolus of 80 units/kg followed by continuous infusion titrated to a target anti-Xa level of 0.3-0.7 IU/mL.

3. Admit to the cardiac telemetry step-down unit for continuous hemodynamic monitoring.

4. Pulmonary embolism response team (PERT) consultation alerted for possible catheter-directed mechanical thrombectomy if hemodynamic deterioration ensues.